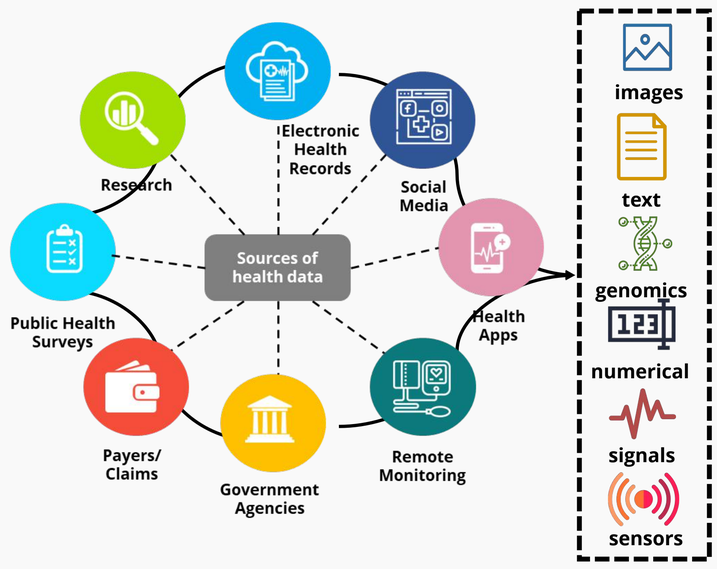
Disruptive digital transformation with
SENSIBLE
(SEquestered aNd Synergistic BLockchain Ecosystem)

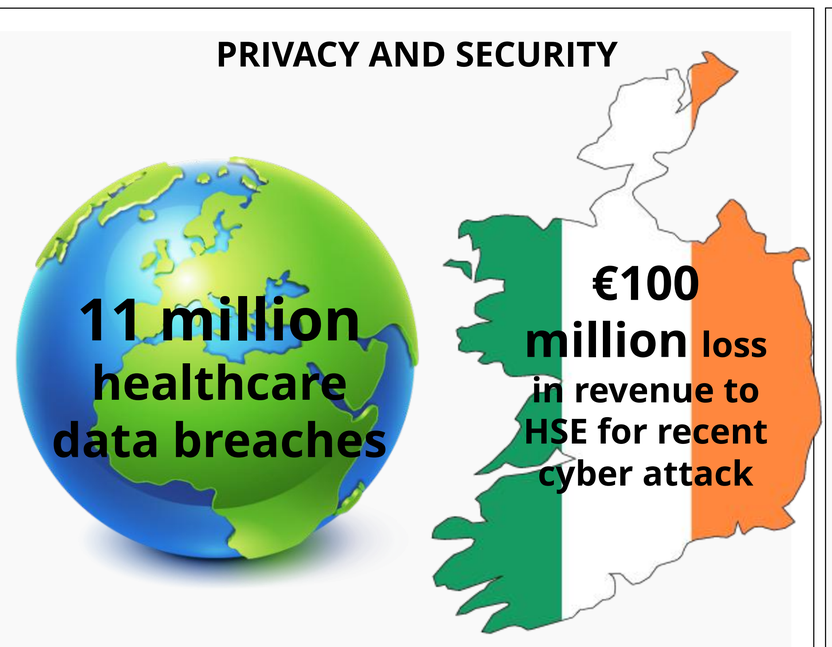
Meghana Kshirsagar[1], Gauri Vaidya[1], Conor Ryan[1]
[1]University of Limerick, Ireland
meghana.kshirsagar@ul.ie, gauri.vaidya@ul.ie, conor.ryan@ul.ie

EXISTING SYSTEMS

MULTIMODAL HEALTH DATA



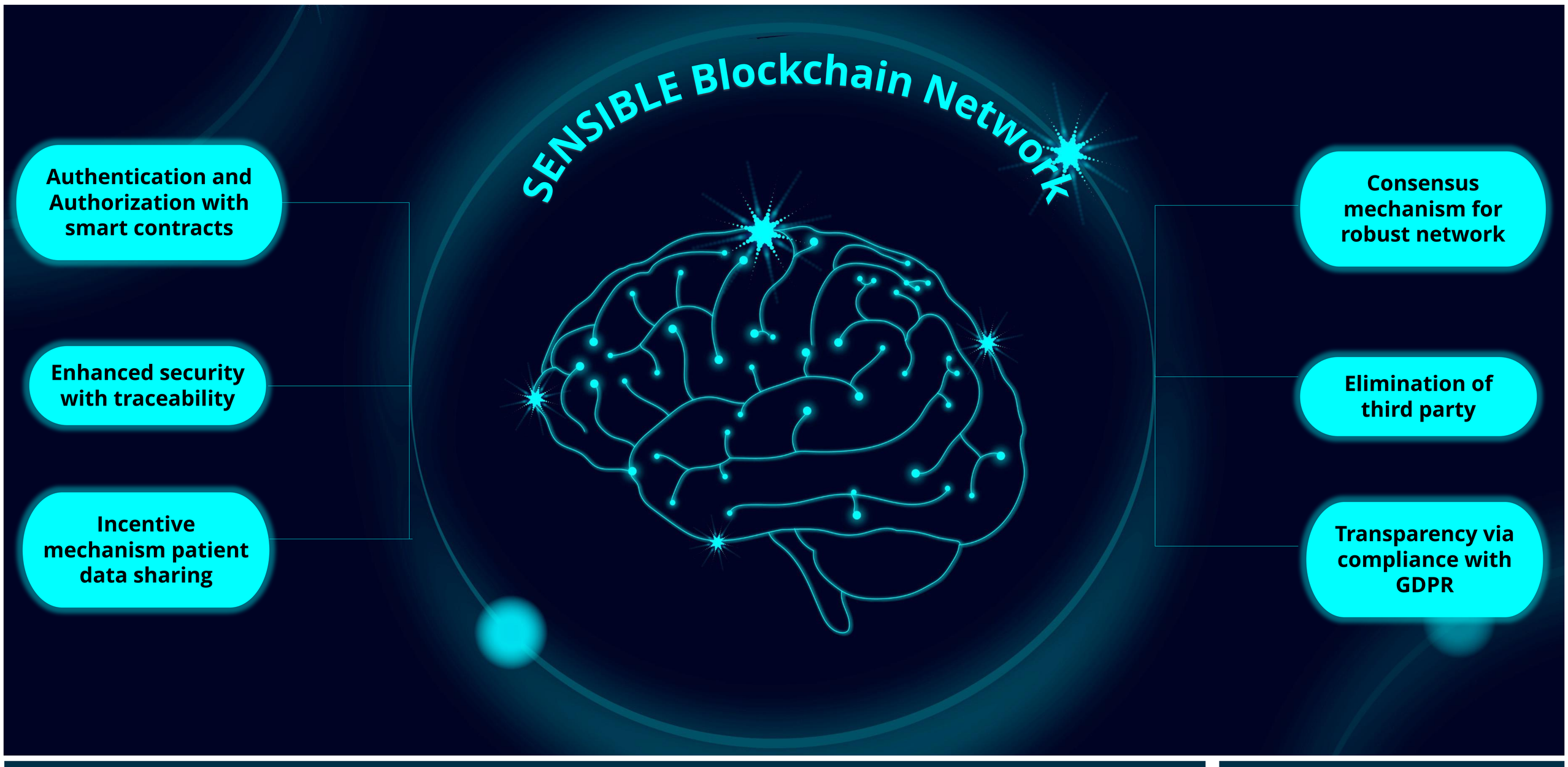
PRIVACY AND SECURITY



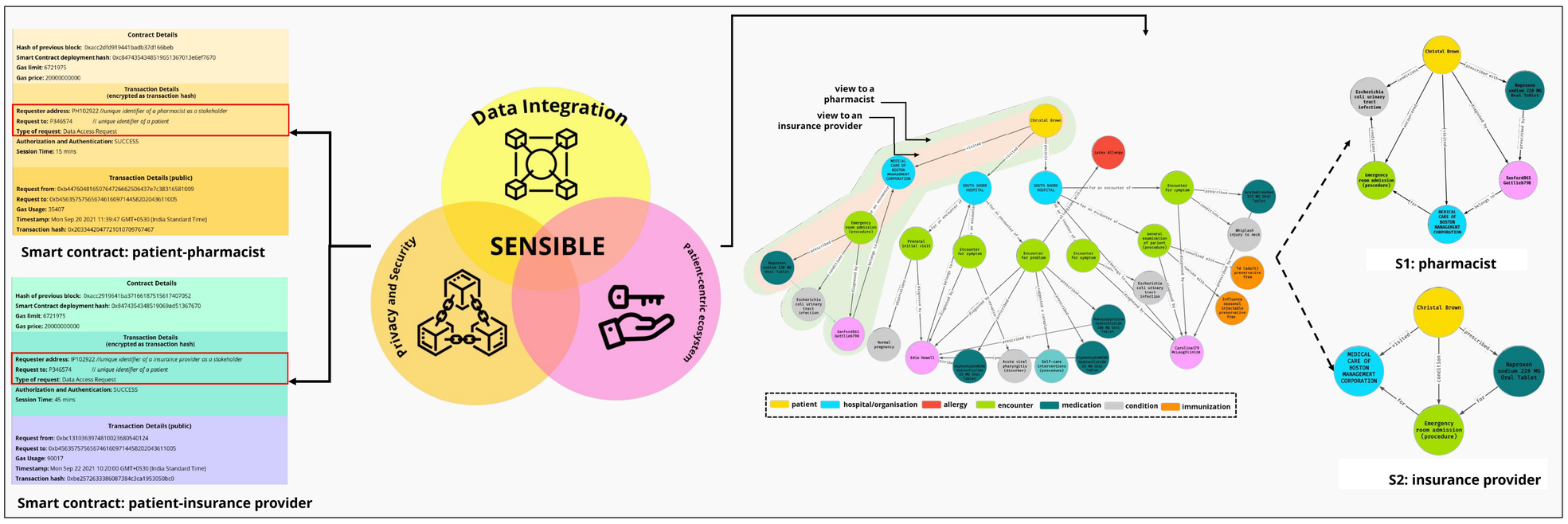
IDEAL WORLD

A ***patient-centric digital healthcare ecosystem*** with:

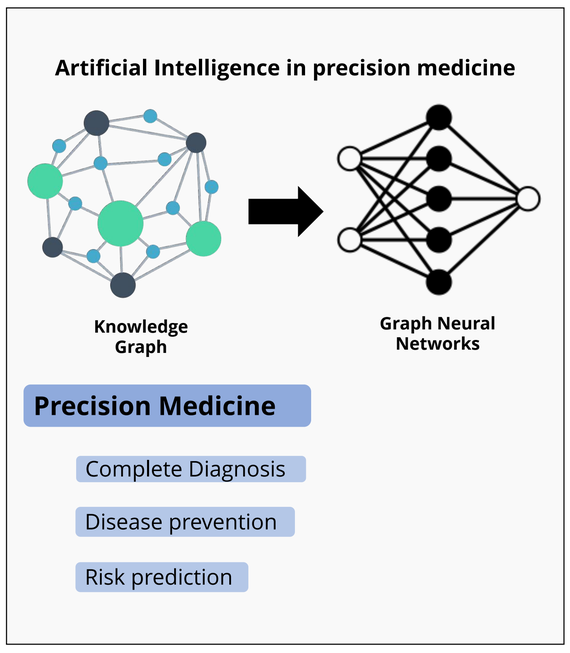
- ***Secured*** and ***seamless*** integration of fragmented health data
- *A **single source of modalities*** from knowledge graphs
- ***Secured consent-based knowledge sharing*** amongst the relevant stakeholders
- ***Personalised healthcare*** via precision medicine



SENSIBLE ECOSYSTEM



MOVING FORWARD



Acknowledgements:
This work was conducted with the financial support of the Science Foundation Ireland Centre for Research Training in Artificial Intelligence under Grant No. 18/CRT/6223

REFERENCES

- Meghana, Kshirsagar Krishn Kumar Gupt, Gauri Vaidya, Conor Ryan, Joseph P. Sullivan, and Vivek Kshirsagar. 2022. "Insights Into Incorporating Trustworthiness and Ethics in AI Systems With Explainable AI." *International Journal of Natural Computing Research* 11 (1): 1–23. https://doi.org/10.4018/IJNCR.310006.
- Yao, Yao, Meghana Kshirsagar, Gauri Vaidya, Jens Ducrée, and Conor Ryan. 2021. "Convergence of Blockchain, Autonomous Agents, and Knowledge Graph to Share Electronic Health Records." *Frontiers in Blockchain* 0 (April): 13. https://doi.org/10.3389/FBLOC.2021.661238.
- Meghana, Kshirsagar, Gauri, Vaidya, Yao, Yao, Smita Kasar, and Conor Ryan. 2022. "SENSIBLE: Sequestered and Synergistic Blockchain Ecosystem". *Wiley engineering Reports (Accepted).* http://dx.doi.org/10.22541/au.165226321.19441546/v1
- Yao, Yao, Meghana, Kshirsagar, Gauri, Vaidya, Junying, Liu, Yongliang, Zhang, and Conor Ryan. 2022. "Blockchain-based Knowledge Graph for High-impact Scientific Collaboration Networks". The Institution of Engineering and Technology. (proof editing)